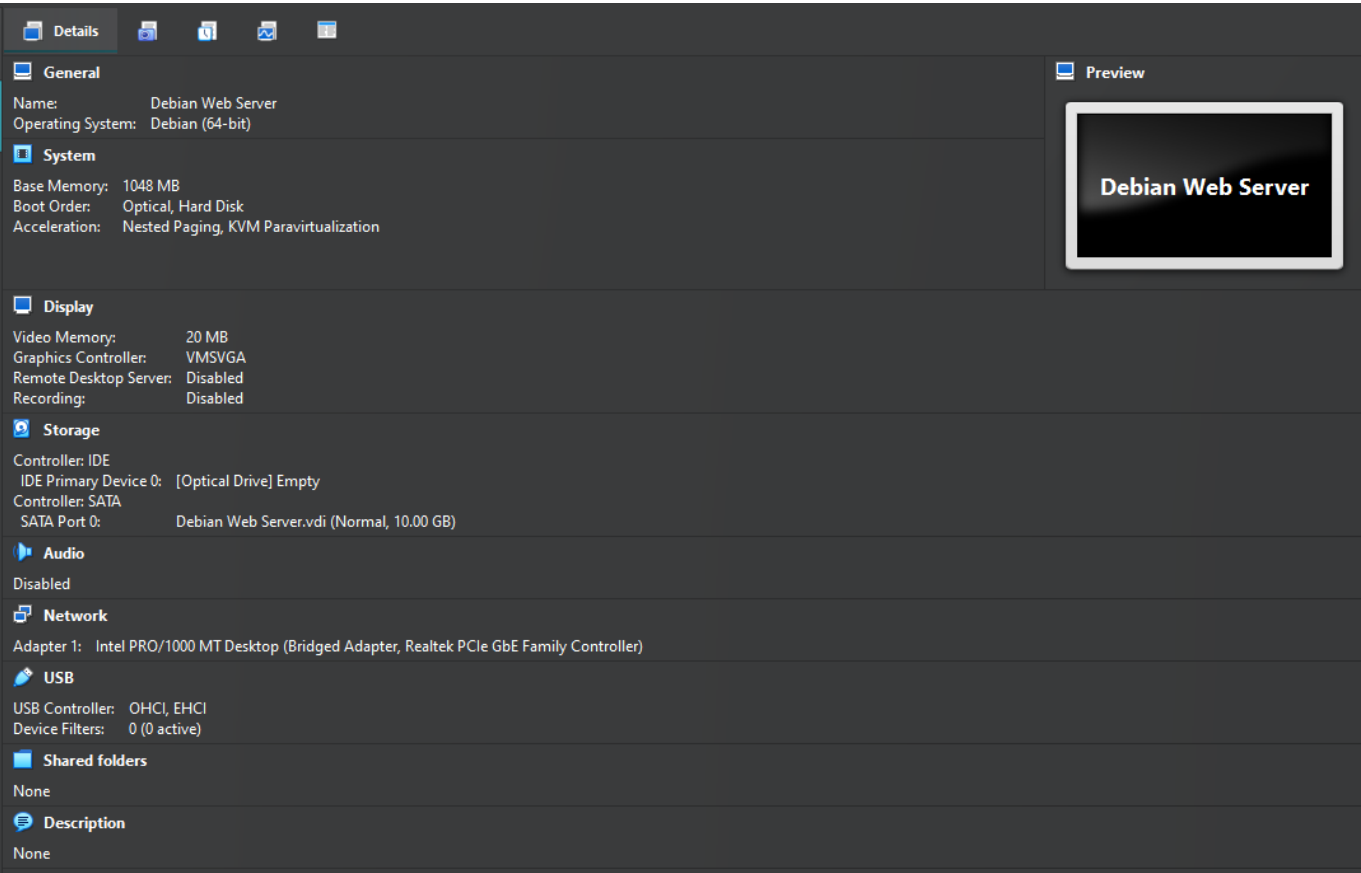


Deliverable 2

1. What are the server hardware specifications (virtual machine settings)?



2. What is the Debian Login Screen?

```
Debian GNU/Linux 13 WebServer tty1
WebServer login: _
```

3. What is the IP address of your Debian Server Virtual Machine?

```
Debian GNU/Linux 13 WebServer tty1
WebServer login: emp
Password:
Linux WebServer 6.12.74+deb13+1-amd64 #1 SMP PREEMPT_DYNAMIC Debian 6.12.74-2 (2026-03-08) x86_64

The programs included with the Debian GNU/Linux system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Debian GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent
permitted by applicable law.
emp@WebServer:~$ hostname -I
192.168.1.121
emp@WebServer:~$ _
```

4. How do you work with the Firewall in Debian? (Type and explain what each command does)

Command name: `ufw`

- **Description:** UFW (Uncomplicated Firewall) is the default and easiest way to manage the firewall in Debian. It simplifies iptables/nftables configuration.
 - **Formula/Syntax:** `sudo ufw [option] [argument]`
 - **Examples:**
 - How do you check if the Firewall is running?
By using the command `sudo ufw status`
 - How do you disable the Firewall?
By using the command `sudo ufw disable`
 - How do you add Apache to the Firewall?
By using the command:
`sudo ufw allow 'Apache'` (HTTP only - port 80)
`sudo ufw allow 'Apache Full'` (HTTP + HTTPS - ports 80 & 443)
-

Additional useful commands

- Detailed status: `sudo ufw status verbose`
- Reload rules: `sudo ufw reload`
- Enable firewall: `sudo ufw enable`

Note: Always allow SSH first with `sudo ufw allow OpenSSH` before enabling the firewall to avoid locking yourself out.

5. What different commands do we use to work with Apache? (Type and explain what each command does and include a screenshot!)

Command name: `apache2 / systemctl`

- **Description:** Apache on Debian uses `apache2` as the service name and `systemctl` to manage it. These commands allow you to control, test, and monitor the Apache web server.
- **Formula/Syntax:** `sudo systemctl [action] apache2` and `sudo apache2ctl [option]`
- **Examples:**
 1. What is the command you use to check if Apache is running?
By using the command `sudo systemctl status apache2`
 2. What is the command you use to stop Apache?
By using the command `sudo systemctl stop apache2`
 3. What is the command you use to restart Apache?
By using the command `sudo systemctl restart apache2`
 4. What is the command used to test Apache configuration?
By using the command `sudo apache2ctl configtest`

5. What is the command used to check the installed version of Apache?

By using the command `apache2 -v`

6. What are some common configuration files for Apache?

- Main configuration: `/etc/apache2/apache2.conf`
- Ports configuration: `/etc/apache2/ports.conf`
- Virtual hosts: `/etc/apache2/sites-available/000-default.conf`
- Enabled sites: `/etc/apache2/sites-enabled/`

7. Where does Apache store logs?

Apache stores logs in: `/var/log/apache2/`

- Access log: `/var/log/apache2/access.log`
- Error log: `/var/log/apache2/error.log`

8. What are some basic commands we can use to review logs?

- View last 50 lines: `sudo tail -n 50 /var/log/apache2/error.log`
- Follow logs in real-time: `sudo tail -f /var/log/apache2/access.log`
- View with less: `sudo less /var/log/apache2/error.log`

```
Debian GNU/Linux 13 WebServer tty1
WebServer login: emp
Password:
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Debian GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent
permitted by applicable law.
emp@WebServer:~$ hostname -I
192.168.1.121
emp@WebServer:~$ sudo systemctl status apache2
[sudo] password for emp:
* apache2.service - The Apache HTTP Server
   Loaded: loaded (/usr/lib/systemd/system/apache2.service; enabled; preset: enabled)
   Active: active (running) since Sun 2026-04-26 19:22:21 EDT; 2min 21s ago
   Invocation: 949c89091e114640ae6e0dae02b0efaf
   Docs: https://httpd.apache.org/docs/2.4/
   Process: 681 ExecStart=/usr/sbin/apachectl start (code=exited, status=0/SUCCESS)
   Main PID: 725 (apache2)
   Tasks: 55 (limit: 1136)
   Memory: 8.4M (peak: 8.7M)
   CPU: 43ms
   CGroup: /system.slice/apache2.service
           └─725 /usr/sbin/apache2 -k start
             └─727 /usr/sbin/apache2 -k start
               └─729 /usr/sbin/apache2 -k start

apr 26 19:22:21 WebServer systemd[1]: Starting apache2.service - The Apache HTTP Server...
apr 26 19:22:21 WebServer apachectl[715]: AH00557: apache2: apr_sockaddr_info_get() failed for WebServer
apr 26 19:22:21 WebServer apachectl[715]: AH00558: apache2: Could not reliably determine the server's fully qualified domain name, using 127.0.0.1. Set the 'ServerName'
apr 26 19:22:21 WebServer systemd[1]: Started apache2.service - The Apache HTTP Server.
lines 1-19/19 (END)
```